Shizuka Takao

shizukat@uci.edu | shizuka-takao.vercel.app | github.com/shizuka-takao

Research Interests: Computer Vision • Reinforcement Learning • Autonomous Systems • ML Systems

EDUCATION

University of California, Irvine — Expected Mar 2027
B.S. Computer Science, Intelligent Systems Specialization | GPA 3.975/4.000

Diablo Valley College — May 2024
A.S.-T. Mathematics; A.S. Computer Science | GPA 4.000/4.000

Mathematics: Calculus I (A), Calculus II (A), Calculus III (A), Linear Algebra (A), Differential Equations (A), Discrete Mathematics (A+)
Machine Learning & AI: Machine Learning (A+), Artificial Intelligence (A+), Graphical Models (A+), Project in AI (A)
Probability: Probability & Statistics for CS (A+), Applications of Probability in CS (A)
Data Structures, Algorithms & IR: Data Structures (A), Information Retrieval (A+)
Systems: Principles of System Design (A-)

RESEARCH EXPERIENCE

Intelligent and Autonomous Systems Lab, UC Irvine — Sep 2026–Present
Undergraduate Research Assistant | WayFinder autonomous drone navigation

- Benchmarked a DQN-based drone controller in AirSim on 1,100 navigation routes across 11 difficulty levels
- Investigating repeated-state loops and stalled goal progress to inform multimodal recovery-waypoint selection

Split Computing for Edge–Cloud AI, UC Irvine — Apr–Jun 2026
PhD student–mentored literature review | Split computing

- Reviewed JointDNN and related split-inference papers on partitioning neural networks across devices and servers
- Discussed model partitioning, feature compression, and latency–bandwidth–energy trade-offs in split inference

Zhang Lab, UC Irvine — Oct–Dec 2025
PhD student–mentored literature review | Spatial transcriptomics

- Reviewed Cell2location and STALocator literature on spatial transcriptomics and single-cell mapping methods
- Explored Scanpy and AnnData workflows, drafted a SLURM batch script, and studied HPC job submission practices

Ding Lab, UC Irvine — Jun 2025–Present
Undergraduate Research Assistant | FISH-ML fluorescence microscopy segmentation

- Extended the existing nucleus-only FISH-ML workflow to Cellpose-SAM-based whole-cell segmentation
- Implemented nucleus–cytoplasm pairing using spatial overlap and Hungarian matching for aligned mask exports
- Identified inconsistent IoU/Dice scoring for unlabeled cells and proposed standardized model comparisons
- Implemented loss masking in Cellpose-SAM training trials; evaluated coverage, nucleus weighting, and diameter
- Investigated low cytoplasm contrast as a cause of poor SAM masks, testing CLAHE and edge enhancement
- Updated existing GUI editing, session, and MATLAB export functions for multichannel whole-cell analysis

INDUSTRY & SPONSORED EXPERIENCE

Gatekeeper Systems — Jan–Mar 2026
Machine Learning Engineering Intern | LLM retrieval and evaluation

- Reviewed Gemini RAG and SQL-generation code, testing query-routing errors and document-page mismatches
- Explored LoRA fine-tuning workflows in Vertex AI with company-specific documentation and example prompts

INDUSTRY & SPONSORED EXPERIENCE (CONTINUED)

Break Through Tech AI Studio – American Express Project Aug–Dec 2025

ML Fellow | Credit-risk classification and ML engineering

- Prepared credit-risk features and addressed class imbalance through preprocessing and supervised model selection
- Trained and evaluated XGBoost classifiers, achieving 0.83 validation accuracy on credit-risk predictions
- Developed a Streamlit app for real-time credit-risk inference using the validated XGBoost classification model
- Led code reviews, merge-conflict resolution, and technical integration for a seven-member project team

Kasanare Corporation Jun–Aug 2023

Software Engineering Intern | Tokyo, Japan

- Prototyped retrieval-augmented chatbot workflows using LangChain and Pinecone for vector search and context prompts

SELECTED AI, ML AND SYSTEMS PROJECTS

Layer-wise Representation Learning for Pet Re-Identification Oct 2026–Present

CS 117 Computer Vision | MegaDescriptor

- Designing layer-wise feature analysis of MegaDescriptor representations for individual dog and cat identity
- Planning identity-retrieval experiments comparing intermediate and final MegaDescriptor network-layer embeddings
- Exploring background and viewpoint robustness of learned identity representations in pet re-identification

Probabilistic Movie Preference Models May–Jun 2026

Naive Bayes, Chow–Liu, Sparse Logistic Regression

- Compared Naive Bayes, Chow–Liu, and L1-regularized sparse logistic regression on 2,000 movie users
- Modeled movie-preference dependencies using mutual information and Chow–Liu maximum spanning trees
- Evaluated preference predictions with 0.7845 accuracy, 0.8390 F1, and 0.4585 log loss across models

Sokoban: Classical Planning and Reinforcement Learning Apr–Jun 2026

PyTorch, Double DQN, A*, Imitation Learning

- Co-developed BFS, Greedy, and A* Sokoban solvers using box–goal matching and deadlock pruning
- Trained an action-masked Double DQN with box-push macro-actions, reward shaping, and curriculum learning
- Used A*-guided imitation learning via behavior cloning; achieved 64% held-out and 51% procedural solve rates

Concurrent Systems Programming Oct–Dec 2025

C, Linux, POSIX Threads

- Implemented concurrent TCP servers using POSIX threads, synchronized work queues, and semaphores
- Built a C heap allocator with segregated free lists, block splitting, coalescing, and robust pointer validation
- Used producer–consumer synchronization and worker threads to coordinate concurrent server requests

HONORS, CERTIFICATIONS AND ACTIVITIES

ICS Honors Program, UC Irvine — selective CS honors and research programme — Aug 2025–Present
Break Through Tech AI Fellow, Cornell Tech — selected from 3,000+ applicants — May 2025–Mar 2026
Machine Learning Foundations Certificate — Break Through Tech AI at Cornell Tech — Aug 2025
S and S International Fund, UC Irvine — $7,500 institutional scholarship for international students — Aug 2025
Secretary, Women in Computer Science — coordinated workshops and speaker events — Sep 2023–May 2024
Mathematics Tutor, Diablo Valley College — tutored through multivariable calculus — Mar–Sep 2023
Inter-Council Representative, Deep Learning Society — supported funding and club initiatives — Jan–May 2023

TECHNICAL SKILLS

Programming: Python, C/C++, Java, SQL, MATLAB | Machine Learning: PyTorch, TensorFlow, XGBoost, Gymnasium

Engineering: OpenCV, GroundingDINO, SAM, Cellpose-SAM, AirSim, Git, Linux, Docker, Kubernetes, AWS